CS Picks: An Uncertainty-Aware Interface for Exploring Computer Science Departments, Faculty, and Venues

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Abstract

Metrics-based rankings of computer science departments have become a routine input to consequential decisions, most visibly for prospective doctoral students choosing where to apply. CSRankings is the field's de facto standard: it is transparent, reproducible, and deliberately narrow, computing a single ordered list from fractional publication credit at a curated set of venues. Its very legibility, however, encourages readers to treat one integer as a measurement of departmental quality.

We present CS Picks, a static, entirely client-side web application that reuses CSRankings's inputs and scoring rules but reorganizes them around the questions an applicant actually asks. CS Picks makes the discretionary parameters of a ranking manipulable and visible, and adds views that a single ordered list cannot express: a *rank stability* sweep that recomputes a department's position under every combination of look-back window and venue set; per-subfield attribution of both departmental score and individual faculty output; a *per-faculty* ordering that measures output per head rather than in aggregate; a *fragility* analysis reporting how many departures would move a department out of its rank band; head-to-head comparison of two departments, two researchers, or two subfields; and a funding view over NSF awards attributed to the same roster.

Applying the stability sweep to all 190 ranked U.S. departments produces our central empirical finding: rank stability is highly unevenly distributed. The median department moves 17 positions across 16 defensible settings, and 77.4% move more than 10, yet the top-10 departments move a median of only 5. The ordering is therefore trustworthy exactly where readers least need guidance and unreliable across the broad middle where most applicants are choosing. We report companion analyses at three levels of granularity — department, researcher, and subfield — including a per-faculty ordering that disagrees with the CSRankings ordering by a median of 13 positions, a subfield distribution in which machine learning alone accounts for 24.0% of all U.S. output, an individual distribution in which the top 10% of researchers produce 40.0% of it, and an NSF funding ordering that agrees with the publication ordering only loosely ($\tau = 0.778$, median displacement 12 positions).

1 Introduction

A prospective doctoral student in computer science faces a decision with a long horizon, poor feedback, and very little trustworthy information. Departmental rankings fill that vacuum. They are compact, ordinal, and easy to act on, and so they are acted on — by applicants, by administrators, and by the departments that manage themselves against them [9, 3].

Within computer science, CSRankings [1] occupies a distinctive position. Unlike reputation surveys, it is computed from a public, auditable procedure over public data: publications recorded in

DBLP [5] at a curated list of selective conferences, credited fractionally among coauthors, aggregated per area, and combined across areas by a geometric mean. Anyone can reproduce the number. This transparency is a genuine advance, and we build directly on it.

Transparency of *procedure*, however, is not the same as transparency of *contingency*. A reproducible pipeline still rests on discretionary choices — which venues count, how many years to look back, whether credit follows an author's current or historical affiliation — and the output format, a single ordered list, cannot express that any of those choices were made. A reader who sees a department at position 32 has no way to learn from the artifact itself that the same department sits at 23 under one defensible setting and 50 under another. The number's precision is real; its stability is not, and nothing on the page distinguishes the two.

A single departmental ordering also answers only one of the three questions an applicant is actually asking. Applicants join research groups, work with individual advisors, and commit to a subfield. CS Picks therefore reports at all three levels — department, subfield, and individual researcher — and lets any two entities of the same kind be placed side by side.

This paper describes CS Picks, a companion tool to a guide for prospective computer science doctoral students [7]. CS Picks is not an alternative ranking. It consumes CSRankings's inputs and reimplements its scoring faithfully, then asks what else can be built once that computation is fast enough to run many times in a browser. Our contributions are:

- **A rank stability sweep** (§3.4) that recomputes a department's rank under all 16 combinations of four look-back windows and four venue sets, holding region fixed, and reports the resulting envelope rather than a point estimate.
- **Per-subfield attribution** (§3.5) of both departmental score composition and individual faculty output, so that a name listed under a subfield reports what that person published *in that subfield* rather than their department-wide total.
- **A per-faculty ordering** (§3.6) that divides adjusted output by publishing headcount, separating how much a department produces from how many people it employs.
- **A fragility analysis** (§3.7) reporting the number of departures that would move a department out of the top 10, 25, or 50, computed without naming the individuals concerned.
- **Head-to-head comparison** (§3.9) of two departments, two researchers, or two subfields, reporting per-area margins and, for departments, a decomposition of the rank gap into the areas that produced it.
- **A funding view** (§3.10) over NSF awards attributed to the same roster, which lets a publication-based ordering be checked against an independently produced one.
- **An empirical characterization** (§6) of rank stability, venue sensitivity, output concentration, per-faculty ordering, fragility, the individual-researcher distribution, the subfield distribution, and funding–publication divergence across the U.S. departments in the current data.
- **A fully client-side implementation** (§4) with no backend, in which a complete re-ranking of 20,822 faculty takes roughly 100 ms, making interactive sensitivity analysis practical.

2 Background

2.1 CSRankings and its scoring rule

CSRankings ranks departments by publication output at a curated set of highly selective conferences, deliberately excluding journals, workshops, and citation counts. Three design decisions matter for everything that follows.

Fractional credit. A paper with k authors contributes $1/k$ to each author, and thence to their department. This is a deliberate counterweight to the very different coauthorship norms across subfields, where raw paper counts would reward large-team areas. We refer to the resulting quantity as *adjusted count* throughout.

Geometric mean across areas. A department's score is

$$S(d) = \left(\prod_{a \in A} (c_{d,a} + 1) \right)^{1/|A|}, \tag{1}$$

where A is the set of top-level research areas and $c_{d,a}$ is department d's adjusted count in area a. The product runs over *all* areas, not only those where the department is active, so a zero contributes a factor of one. This strongly rewards breadth: a department strong in every area beats one that is dominant in a few. It is the single most consequential choice in the formula and is often misread as an average of strengths. §6.8 shows it operating on a concrete pair of departments whose total output differs by 0.3%.

Venue curation. Which conferences count is a maintained, contested list. CSRankings distinguishes a default set from a larger set including additional venues; independent efforts such as the CORE rankings [2] draw the line elsewhere. CS Picks supports four sets (§3.3).

An important structural consequence of Eq. 1 is that scores are rounded to one decimal place before ranking, and ties share a rank under standard competition ranking. In the tail of the distribution, where departments differ by fractions of a publication, this rounding produces substantial tied blocks — a fact that becomes visible in our stability analysis.

2.2 Why a single ordered list under-serves the applicant

The critiques of academic ranking are well rehearsed: rankings reshape the behavior they claim to measure [10], and institutions reorganize around them [9]. Our concern is narrower and more practical. Even granting that adjusted publication count at selective venues is a reasonable proxy for research activity, the *presentation* of that proxy as a single integer discards three things an applicant needs:

1. **Sensitivity.** How much of position 32 is the department, and how much is the window and venue set?
2. **Locality.** An applicant does not join a department; they join a research group. Departmental aggregates say little about whether one subfield is strong or whether one professor is taking students.
3. **Scale.** A sum over faculty conflates a productive department with a large one, and says nothing about how much of the total rests on how few people.

Table 1: The corpus as fetched at the time of writing. Publication records are per author, area, and year; they are aggregates, not individual papers. The final block covers the separately synchronized NSF award snapshot (§3.10).

Quantity	Value
Faculty in roster	20,822
Institutions	702
Publication records	296,267
Year range	1970–2026
Faculty with a homepage	20,822 (100.0%)
Faculty with a Scholar ID	18,256 (87.7%)
NSF awards in snapshot	22,696
within the 2016–2026 window	10,226
Faculty matched to an award	4,101
Institutions matched	182
Attributed award value	\$4.12B

CS Picks addresses each in turn.

3 Methods

3.1 Data sources and joins

CS Picks fetches three CSRankings CSV files at page load — the faculty roster (name, affiliation, homepage, Scholar and ORCID identifiers), the per-author publication table (name, area, year, raw count, adjusted count), and the institution table (region, country, homepage) — and joins them on author name. Table 1 summarizes the resulting corpus.

A consequence of the third row deserves emphasis, because it constrains every method below: the publication table is *aggregated per author, area, and year*. It contains no paper titles, no paper identifiers, and no author order. Any analysis requiring per-paper structure — coauthorship, first-versus-last authorship, cross-institutional collaboration — cannot be answered from CSRankings data alone and must go to DBLP directly, one author at a time.

Two optional sources extend the corpus. Affiliation histories derived from OpenAlex [8] allow publications to be credited to the institution an author held *at publication time* rather than their current one; these are generated offline and loaded on demand. NSF award records [6], synchronized offline and attributed fractionally among principal investigators, support a separate funding view (§3.10).

3.2 The filtering and ranking pipeline

The core computation, invoked by nearly every view, is a three-stage pipeline parameterized by a year range $[y_0, y_1]$, a region r, a venue set V, and optionally an affiliation history map H:

1. **Collect.** Retain publications with $y_0 \leq \text{year} \leq y_1$ whose venue is in V; credit each to the institution(s) implied by H (or the current affiliation when H is absent), keeping only those in r. Accumulate per-department, per-area, and per-faculty totals in the same pass.

2. **Score.** Apply Eq. 1 to each department.

3. **Rank.** Sort by score and assign standard competition ranks, overall and within each area, so tied scores share a rank and the next distinct score skips ahead.

Crucially, per-faculty and per-area-per-faculty totals are accumulated *during* stage 1 rather than derived afterwards. Under historical attribution a single author's output may be split across several institutions, and that split cannot be reconstructed from their aggregate area totals once the pass has finished. The same accumulation is what makes the fragility analysis (§3.7) and the per-subfield researcher tables (§6.6) computable at all.

3.3 Venue sets

CS Picks supports four venue sets: CSRankings's default list; CSRankings's extended list including additional venues; CORE A* only; and CORE A together with A*. A venue that upstream scrapes but assigns to no research area is counted by none of them, matching CSRankings's own behavior.

3.4 Rank stability

Let $W = \{5, 10, 20, 30\}$ be look-back windows in years and V the four venue sets. For a fixed region r and end year y_1, the sweep evaluates the pipeline at every $(w, v) \in W \times V$, yielding 16 complete rankings. For a department d we report the envelope

$$\Big[\min_{(w,v)} \rho_{w,v}(d),\ \max_{(w,v)} \rho_{w,v}(d)\Big], \tag{2}$$

its median, and the *spread* $\max - \min$, where $\rho_{w,v}(d)$ is d's rank in the corresponding run.

Two design decisions are worth stating explicitly.

Region is held fixed. A rank among U.S. departments and a rank worldwide answer different questions over different populations; pooling them into a single interval would mix a *scope* choice into what is meant to be a range over *methodological* choices. The user's selected region is therefore held constant across the sweep.

Departments that vanish are reported, not dropped. A department active only at venues outside a given set does not appear in that run's ranking at all. Silently omitting such runs would understate volatility, so we report the count of settings under which a department is unranked alongside its envelope.

One sweep produces ranks for every department simultaneously, so its cost is amortized: the result is cached per region and reused for any department the user subsequently inspects.

3.5 Per-subfield attribution

Departmental cards list each active subfield with the faculty contributing to it. Two quantities are distinguished that a naive implementation conflates. First, a faculty member listed under subfield a reports their output *within a at that institution*, not their department-wide total — which, for a broad researcher, may differ by an order of magnitude. Second, we report each subfield's share of the department's own score,

$$\sigma_{d,a} = \frac{\ln(c_{d,a} + 1)}{\sum_{a' \in A} \ln(c_{d,a'} + 1)}, \tag{3}$$

which follows from taking logarithms of Eq. 1. Because these shares are internal to one department and sum to one, they describe composition, not standing: a large $\sigma_{d,a}$ means area a drives d's score, not that d leads others in a. The interface states this explicitly, because the distinction is easy to lose.

3.6 Per-faculty ordering

Eq. 1 accumulates adjusted counts and then takes a geometric mean across areas. Because the per-area counts are sums over faculty, a department can raise its score by employing more people at a constant rate of output. The resulting order therefore answers "how much does this department publish", which is not the same question as "how much does a member of this department publish" — and for an applicant, who will work with a handful of people rather than the whole department, the second is at least as relevant.

We therefore also order departments by

$$P(d) = \frac{\sum_{a \in A} c_{d,a}}{|F_d|}, \tag{4}$$

where F_d is the set of faculty at d with at least one credited publication in the selected window. Headcount is measured as *publishing* faculty rather than roster size, so that a department is not penalized for listing members whose work falls outside the counted venues.

Departments with fewer than five publishing faculty are excluded from this ordering rather than ranked. With two or three people the ratio is substantially one person's output, and the resulting list is dominated by small groups rather than by productive ones. This threshold is a judgement, and we report it explicitly because it materially determines the head of the ordering.

3.7 Fragility

Rank stability (§3.4) asks how much of a position is an artifact of methodology. Fragility asks a complementary question about personnel: how many departures would move this department out of its rank band?

Because removing faculty from one department leaves every other department's score untouched, a hypothetical rank is a lookup rather than a re-ranking. We proceed greedily. At each step we evaluate removing each remaining faculty member, recompute the department's score under Eq. 1 from its per-area counts less that person's per-area contribution, and remove whichever costs the department most. The choice is not simply the most prolific member: under a geometric mean, losing the only contributor in a thin area can cost more than losing a larger producer in a crowded one. We report the number of departures needed to leave the top 10, 25, and 50, and the rank trajectory of the first several steps.

Computing this requires per-person, per-area credit, which the pipeline accumulates during stage 1 (§3) rather than deriving afterwards.

Names are deliberately withheld. The procedure necessarily identifies which individuals carry a department, and the interface does not display them. Publishing a ranked list of the people whose departure would most damage their employer invites precisely the personnel conclusions this work should not support, and the aggregate quantity — how concentrated a department's output is — is what carries the analytic content. We report distributions across departments in §6 for the same reason, and §8 explains why the researcher tables of §6.6 are treated differently.

3.8 Researcher and subfield metrics

The same pipeline produces per-researcher and per-subfield aggregates, since stage 1 accumulates both. For a researcher we report adjusted count, paper count, *breadth* (the number of top-level areas with any credit), *primary-area share* (the fraction of adjusted output in their largest area), and a normalized-entropy *balance* over the area vector. Two temporal quantities are computed over the selected window: *consistency*, the fraction of years with at least one credited publication, and *momentum*, the change in adjusted output between the trailing three-year window and the three years before it. We call a researcher a *specialist* when primary-area share is at least 60%, and a *generalist* when breadth is at least 4; the thresholds are reporting conventions, not claims.

For a subfield we report region-wide adjusted total, the number of departments and distinct researchers with any credit, growth against the equal-length preceding period, and the share of the subfield's national output held by its five largest departments.

3.9 Head-to-head comparison

Two entities of the same kind are compared by taking the union of their active areas, ordering it by combined credit, and reporting the per-area margin $c_{A,a} - c_{B,a}$. Alongside it we report a small scoreboard (rank, papers, adjusted count, publishing faculty, and areas led, for departments; affiliation, papers, adjusted count, active areas, and areas led, for researchers).

For departments we additionally decompose the score gap. Taking logarithms of Eq. 1, the difference in scores is a sum of per-area terms

$$\ln S(A) - \ln S(B) = \frac{1}{|A|} \sum_a \big[\ln(c_{A,a} + 1) - \ln(c_{B,a} + 1) \big], \tag{5}$$

so ranking areas by $|\ln(c_{A,a} + 1) - \ln(c_{B,a} + 1)|$ identifies exactly the areas that produced the rank gap. This is worth stating because the answer is frequently counterintuitive: the area where the raw margin is largest is usually *not* the area that moved the rank, since the logarithm compresses differences at high volume and magnifies them near zero. §6.8 gives a case where the two answers point in opposite directions.

Subfields are compared by a separate procedure, reporting each side's region-wide total, growth, participating departments and researchers, and the researchers and departments active in both.

3.10 NSF award attribution

The funding view is built from a snapshot of the NSF award database [6], synchronized offline by querying the award-search API once per (faculty, institution) pair over the CSRankings roster. Each award's value is split equally among its listed investigators,

$$\text{share}(p, w) = \frac{\text{value}(w)}{|\,\text{investigators}(w)\,|}, \tag{6}$$

and a person's attributed total is the sum of their shares. Value is the estimated total award amount where present and the obligated amount otherwise. An institution's attributed total is the sum over its matched faculty.

Equal splitting is a deliberate simplification and the main source of error in this view: NSF publishes no per-investigator allocation, so a weighted split would require inventing weights. We therefore report the equal share as a *share*, and separately report the full project value, which is the more appropriate number for judging the scale of the work a group participates in. The two differ substantially for large collaborative awards, and the interface shows both.

A name-reconciliation step maps award investigators onto the current CSRankings roster. CSRankings spells some faculty differently in its roster file than in its publication table, and only names verified against the publication table are retained, so an award with no resolvable investigator is dropped rather than attributed to an approximate match.

4 Implementation

4.1 Architecture

CS Picks is a static, multi-page web application with no backend of any kind. All data is fetched client-side from public sources and processed in the browser; deployment is a directory of static files. This is a substantive design choice rather than an economy: it means the computation a reader sees is the computation they can inspect, there is no server-side state that could personalize or silently alter a ranking, and the tool cannot accumulate a record of who looked up whom.

The application comprises four pages — search and analysis, a ranking-impact simulator, a discoveries feed, and an NSF funding explorer — each with its own entry module, sharing a common filter bar, chart layer, and autocomplete component. The canonical data load is memoized, so modules loaded together share a single download and parse.

4.2 Performance

A complete pass of the pipeline over all 20,822 faculty and 296,267 publication records — filter, score, and rank — takes approximately 100 ms. This number is what makes the stability sweep viable: 16 full re-rankings complete in about 1.7 s.

Because that much uninterrupted computation would freeze the interface, the sweep yields to the event loop between runs and reports incremental progress, and its result is cached per region. The cost is therefore paid once per region per session, and every department examined afterwards resolves from cache.

4.3 Offline generators and correctness

Two datasets are too large or too slow to build at page load and are generated offline: OpenAlex-derived affiliation histories, and the NSF award snapshot.

Correctness is maintained by a unit suite covering the filtering, scoring, and ranking logic, rendering safety, and the metric definitions introduced here, together with an end-to-end browser suite exercising the principal user flows. Every number reported in §6 is emitted by a script that imports the shipping modules and runs them against the live CSRankings inputs, so the paper cannot drift from the implementation without the discrepancy being visible on the next regeneration.

5 Web interface

5.1 Search and filtering

A single search box resolves institutions, faculty, research areas, and individual venues, disambiguating by type. A shared filter bar exposes the discretionary parameters — region, year range, venue set, whether to show rank ordinals, and whether to use historical affiliation — and persists them across pages and into the URL, so any view is shareable as a link that encodes the settings that produced it. Making these parameters ambient rather than buried is itself an argument: the reader is continuously reminded that the numbers are conditional.

5.2 Departmental cards

A department expands to show, in order: the complete faculty roster ordered by adjusted count, with each member's rank within the department when rank display is enabled; each active subfield with its contributing faculty and their *within-subfield* output; and the subfield share of the department's own score (§3.5) rendered as a ranked bar chart.

The bars in that chart are scaled against the largest share rather than against 100%. Because the shares derive from $\ln(c+1)$, the largest is typically under 8%, and drawing each bar at its literal percentage of the available width renders two dozen visually indistinguishable stubs. This is a small detail with a real consequence: the corrected scaling is what makes the composition legible at a glance.

5.3 Analysis panel

Selecting a department or researcher opens a tabbed panel: activity, faculty diversity, publishing effort, conference trends, area growth, collaboration proxies, and rank stability, with the fragility analysis of §3.7 beneath it. Tabs that do not apply to the current target type are hidden rather than shown empty — a researcher has no faculty diversity — and the researcher variant of the activity tab reports the pattern metrics of §3.8 instead of a departmental rank trajectory.

The default view, shown before any search, ranks departments and researchers in two columns. A control above it switches the departmental column between the CSRankings score and the per-faculty ordering of §3.6; the control appears only in that view, since a filtered result list is not a ranking.

5.4 Comparison

A query of the form `A vs B` switches into a head-to-head view. The autocomplete completes only the trailing side, so a comparison can be typed without retyping the first entity, and the same syntax works for two departments, two researchers, or two subfields; mixed-type comparisons are refused with an explanation rather than silently coerced. The view renders the scoreboard, the per-area margin chart, and — for departments — the rank-gap decomposition of Eq. 5. Worked examples appear in §6.8.

5.5 Simulator and discoveries

A separate simulator estimates how adding, removing, or transferring faculty would change a department's rank, resolving hypothetical candidates against DBLP so that real researchers not yet in CSRankings can be evaluated. Because removing faculty leaves every other department's score fixed, the recomputation is a rescore of the affected departments plus a re-sort, and runs interactively.

A discoveries page surfaces notable movements without requiring the reader to know what to search for: rank climbers and decliners, output and breadth changes, subfield growth and consolidation, changes of subfield leadership, and — for U.S. and worldwide scope — the funding sections of §6.9.

5.6 Funding search

The funding page searches the NSF snapshot independently of the publication data; the search page never loads it. Its autocomplete offers three groups — universities, principal investigators, and NSF program names — so `Secure & Trustworthy Cyberspace` is as valid a query as `Carnegie Mellon University`. Results render as two parallel lists, one of institutions and one of faculty, each showing

Table 2: Rank spread (worst minus best position) across 16 settings — four look-back windows × four venue sets — for 190 ranked U.S. departments. Strata are defined by baseline rank.

Stratum	Departments	Median spread
Baseline rank ≤ 10	10	5.0
Baseline rank ≤ 25	26	6.0
Baseline rank ≤ 50	51	10.0
All ranked departments	190	17.0
Distribution over all departments		
First quartile of spread		11 positions
Median spread		17 positions
Third quartile of spread		25 positions
Maximum spread		46 positions
Departments moving >10		77.4%
Departments moving >25		23.2%

award count, attributed share, and full project value. The `A vs B` syntax works here too, over either institutions or investigators.

A permanently visible data-health panel reports snapshot freshness, roster coverage, and the matching caveats of §3.10, because a funding figure invites more precise interpretation than it can support.

6 Features and results

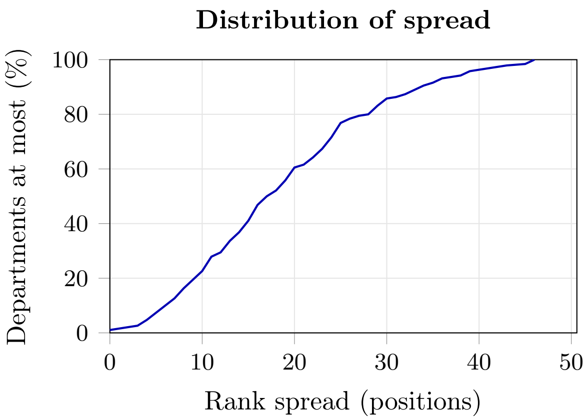
All figures in this section were computed by a script that runs the shipping implementation against the live CSRankings inputs; none are estimated or illustrative. Unless noted, the region is the United States, the end year is 2026, and the baseline setting is a 10-year window with CSRankings's default venue set. Institution names are abbreviated as the interface abbreviates them.

6.1 Rank stability is unevenly distributed

Table 2 summarizes the sweep over all 190 ranked U.S. departments.

The headline number is the median spread of 17 positions, and the fact that over three quarters of departments move by more than 10. But the stratification is the more interesting result: the top of the ranking is markedly more stable than the middle. Departments with a baseline rank of 10 or better move a median of 5 positions; the field as a whole moves a median of 17. Figure 1 shows the full picture — the left panel gives the cumulative distribution of spread, the right panel plots each department's spread against its baseline rank, where the widening cone is the stratification result rendered directly.

The ordering is thus most trustworthy precisely where it is least decision- relevant. An applicant weighing the top handful of departments is looking at a comparatively robust ordering — and is also choosing among options that a single ranking position barely distinguishes. An applicant weighing options ranked between 30 and 80, which describes the large majority of applicants, is reading an ordering in which a 20-position difference can be produced entirely by moving the window from 10 years to 20.



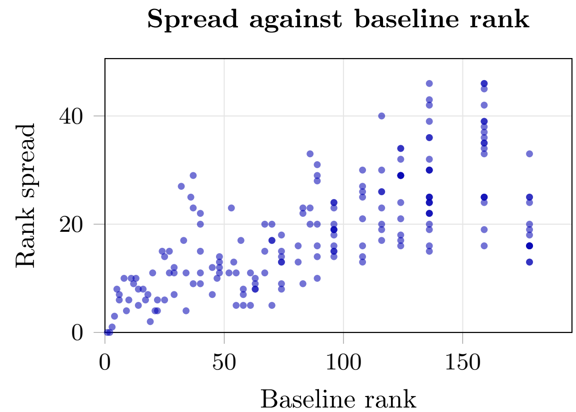


Figure 1: Rank spread across the 16-setting sweep, over 190 ranked U.S. departments. Stability is not a property of the ranking; it is a property of where in the ranking a department sits.

Table 3: Rank envelopes for selected departments across the 16 settings.

Department	Baseline	Best	Worst	Median	Spread
CMU	1	1	1	1	0
UIUC	2	2	2	2	0
Northeastern	11	8	18	11	10
GMU	32	23	50	42	27

Even at the top, stability is not absolute: of the 10 departments in the baseline top 10, only 5 remain in the top 10 under all 16 settings.

Table 3 gives individual envelopes.

CMU and UIUC are genuinely invariant: they hold their positions under every setting tested. GMU ranges from 23rd to 50th depending on nothing but the window and venue set, and its *median* position across the sweep is 42 — ten places below the baseline value a reader would see. So the baseline is not merely imprecise here; it is optimistic relative to the methodology's own central tendency. Reporting only 32 conveys a precision the data does not support, and this is the ordinary case, not a pathological one.

6.2 Venue sets correlate globally but move individual departments

Table 4 compares venue sets pairwise at a fixed 10-year window.

Read as aggregate correlations, these numbers are reassuring: τ never falls below 0.95, so the venue sets broadly agree about the shape of the field. Read as individual outcomes, they are not. Switching from CSRankings's default set to CORE A* moves the median department 5 positions and some department 20 positions, and changes the membership of the top 10.

Table 5 makes the last point concrete by printing all four top-tens side by side. These are the same 190 departments, the same 10-year window, and the same scoring formula; only the venue list differs.

Four departments — CMU, UIUC, UCSD, and Georgia Tech (except under CORE A*, where it falls to 5) — are effectively fixed. Below them the list is unsettled. Stanford is 10th under the default set and under CORE A* but appears in neither of the other two top tens. Northeastern

Table 4: Pairwise agreement between venue sets at a 10-year window. τ is Kendall's rank correlation [4]; overlap is the number of departments common to both top-10 lists.

Set A	Set B	τ	Median $\lvert\Delta\text{rank}\rvert$	Max $\lvert\Delta\text{rank}\rvert$	Top-10 overlap
CSRankings default	CSRankings all	0.982	2	18	9
CSRankings default	CORE A*	0.953	5	20	9
CSRankings default	CORE A/A*	0.973	2	20	9
CSRankings all	CORE A*	0.952	5	19	8
CSRankings all	CORE A/A*	0.973	3	15	10
CORE A*	CORE A/A*	0.961	4	15	8

Table 5: The U.S. top ten under each of the four venue sets, 10-year window. Ties share a rank, so a column may list eleven names. Departments appearing in some columns but not others are set in italics.

	CSRankings default	CSRankings all	CORE A*	CORE A/A*
1	CMU	CMU	CMU	CMU
2	UIUC	UIUC	UIUC	UIUC
3	UCSD	UCSD	UCSD	UCSD
4	Georgia Tech	Georgia Tech	MIT	Georgia Tech
5	MIT	UW & MIT	Georgia Tech	MIT
6	UW & Berkeley	—	UMich	UMD
7	—	UMD	UC Berkeley	UMich
8	UMD	UC Berkeley	*Cornell* & UW	UW
9	UMich	Northeastern	—	UC Berkeley
10	Stanford	UMich	Stanford & *Purdue*	*Northeastern*

is 9th under CSRankings's full venue list and 10th under CORE A/A*, but is outside the top ten under the default set and under CORE A*. Cornell and Purdue appear only under CORE A*. A reader who has internalized "the top ten" as a stable object is reading an artifact of one venue list.

6.2.1 What CSRankings and CORE A* actually disagree about

The rank movement above has a cause that is visible in the venue lists themselves, before any ranking is computed. Table 6 sets the two lists side by side per area.

The headline numbers are close — CSRankings's full list carries 77 venues against CORE A*'s 58 — but the overlap is only 47. Thirty CSRankings venues are absent from CORE A* and eleven CORE A* venues are absent from CSRankings, so the two lists differ on 41 venues, over a third of their union. The two agree exactly on only five of 27 areas: computer vision, information retrieval, security, databases, and logic and verification.

Two structural differences do most of the work. First, CORE A* is broader in AI and theory — it adds AAMAS, KR, and ICAPS to AI, COLT and ICDM to machine learning, and PODC to algorithms — which is why the AI-heavy departments in Table 5 hold their positions under it. Second, and far more consequentially, *CORE A* has no venue at all in seven of the 27 areas*: HPC, robotics, design automation, cryptography, computational biology, visualization, and CS education. CSRankings counts 15 venues across those areas (SC, ICRA, IROS, RSS, DAC, ICCAD, CRYPTO, EUROCRYPT, and others); CORE A* counts none.

Under Eq. 1 an area with no venues is not merely unmeasured, it is inert: $c_{d,a} = 0$ for every

Table 6: Venue counts per area under each set, with the venues on which CSRankings (either list) and CORE A* disagree. Areas are ordered by how much the two disagree; the five areas where they agree exactly are omitted. Venues marked † are in CSRankings's extended list but not its default one.

	Venue count				Venues in one list only	
Area	Def.	All	A*	A/A*	CSRankings only	CORE A* only
AI	2	2	5	10	—	aamas, kr, icaps
HPC	3	3	0	3	sc, hpdc, ics	—
Mobile Computing	3	3	4	4	mobisys	percom, ipsn
Operating Systems	2	5	2	9	eurosys†, fast†, usenixatc†	—
Graphics	2	3	5	7	eurographics†	acmmm, ismar
Robotics	3	3	0	1	icra, iros, rss	—
Machine Learning	3	4	6	8	—	colt, icdm
Design Automation	2	2	0	4	dac, iccad	—
Embedded & Real-Time	3	3	1	2	emsoft, rtas	—
Programming Languages	2	4	2	6	oopsla†, icfp†	—
Networks	2	2	2	3	nsdi	infocom
Cryptography	2	2	0	3	crypto, eurocrypt	—
Comp. Biology	2	2	0	1	ismb, recomb	—
NLP	3	3	2	5	naacl	—
Architecture	3	4	3	4	micro	—
Measurement	2	2	1	2	imc	—
Software Engineering	2	4	3	15	issta†	—
Algorithms	3	3	4	11	—	podc
Econ & Computation	2	2	1	1	wine	—
HCI	3	3	2	6	ubicomp	—
Visualization	2	2	0	1	vis	—
CS Education	1	1	0	4	sigcse	—
Total venues	**64**	**77**	**58**	**148**	30	11

department, so the factor $(c_{d,a} + 1)$ equals 1 everywhere and the area stops discriminating between departments entirely. Switching to CORE A* therefore does not reweight seven areas, it deletes them, and the geometric mean silently becomes an average over 20 effective areas rather than 27.

This is measurable in exactly the way it should be. Across the 188 departments ranked under both sets, the median department takes 16.8% of its adjusted output in those seven areas, and a department's exposure to them predicts how far it falls: the correlation between dropped-area share and rank shift is $r = 0.495$. The 72 departments with at least 20% of their output in the dropped areas move a median of 2 positions worse under CORE A*, while the 50 departments with under 10% move a median of 7 positions better — a nine-position swing driven by nothing but which subfields a reader's chosen venue list happens to cover.

For an applicant in robotics or cryptography the practical reading is blunt: a CORE A* ranking contains no information about their subfield whatsoever, and a department's position under it reflects only the work it does elsewhere. This is the strongest argument in the paper for exposing the venue set as a visible control rather than a buried constant.

Table 7 reports the departments that CORE A* treats most differently from the CSRankings default.

Two things stand out. First, the largest movers cluster in the tail, where the tied blocks noted in §2 mean a small score change crosses many departments at once: the five departments tied at 136

Table 7: Departments moved most by switching from CSRankings's default venue set to CORE A*, 10-year window. Negative shift means the department ranks better under CORE A*.

Department	CSRankings default	CSRankings all	CORE A*	CORE A/A*	Shift
*Gains under CORE A**					
Auburn	116	113	101	105	−15
Brigham Young	116	113	101	115	−15
Cincinnati	116	120	101	115	−15
Drexel	89	89	75	84	−14
Emory	83	82	71	75	−12
UC Santa Cruz	37	38	27	30	−10
*Losses under CORE A**					
Memphis	108	113	128	115	+20
Washington State	86	86	101	87	+15
Delaware	70	76	85	75	+15
Florida Atlantic	136	141	150	156	+14
Northern Arizona	136	141	150	138	+14
Alabama	136	141	150	138	+14

Table 8: Concentration of adjusted publication count, for 166 U.S. departments with at least 5 active faculty in a 10-year window.

Quantity	Value
Median faculty per department	24
Median top-1 share	15.2%
Median top-3 share	36.7%
Departments with top-3 share > 50%	31.9%
Median top-3 share, rank ≤ 25	15.5%
Median top-3 share, rank > 100	60.3%

under the default set move together. Second, the movement is not confined to the tail. UC Santa Cruz sits at 37 under the default set and 27 under CORE A* — a ten-place move within the range applicants actually compare, produced entirely by substituting one defensible venue list for another.

This is the familiar gap between a distribution and a case. A department evaluating itself, or an applicant evaluating a department, experiences the 20-position move, not the 0.953 correlation. High rank correlation is compatible with substantial individual displacement, and reporting only the former is how sensitivity analyses come to reassure when they should caution.

6.3 Output concentration rises sharply down the ranking

Table 8 reports the share of a department's adjusted output produced by its most prolific faculty, over 166 U.S. departments with at least five active faculty.

Nearly a third of departments derive more than half their adjusted output from three people, and the contrast across the ranking is stark: a median top-3 share of 15.5% among the top 25 versus 60.3% below rank 100. Departments in the tail are not scaled-down versions of departments at the top; they are structurally different, with output concentrated in a few individuals.

For an applicant this is arguably more actionable than the rank itself. A department ranked 120

Table 9: The two departmental orderings side by side, 10-year window, over the 166 U.S. departments with at least five publishing faculty. The left block orders by the CSRankings score of Eq. 1 and reports where each department sits per faculty; the right block orders by adjusted output per publishing faculty member and reports where each sits in the CSRankings order.

	Ordered by CSRankings score			Ordered per faculty			
#	Department	Score	Per-fac.	Department	Per fac.	Faculty	CSRankings
1	CMU	20.6	7	Stanford	11.66	61	10
2	UIUC	16.7	10	UC Berkeley	8.32	85	7
3	UCSD	14.8	14	UCLA	7.55	50	20
4	Georgia Tech	13.7	30	MIT	7.50	104	5
5	MIT	11.3	4	UT Austin	7.28	60	14
6	UW	10.7	6	UW	7.04	80	6
7	UC Berkeley	10.6	2	CMU	6.59	183	1
8	UMD	10.3	12	Penn	6.31	70	18
9	UMich	10.1	23	Harvard	6.22	38	39
10	Stanford	9.6	1	UIUC	6.16	127	2
11	Northeastern	9.4	39	Princeton	5.90	67	17
12	Purdue	8.5	29	UMD	5.88	98	8
13	Cornell	8.2	19	Columbia	5.83	58	19
14	NYU	7.7	17	UCSD	5.62	130	3
15	UT Austin	7.7	5	UNC	5.58	46	43

whose standing rests on three faculty is a good choice if one of those three takes the applicant as a student and a poor one otherwise, and the rank alone cannot distinguish these outcomes. It also bears on the stability result above: concentrated departments are precisely those whose position is most sensitive to which venues are counted, since a few individuals' venue preferences determine the department's profile.

6.4 Measuring per faculty reorders the field

Table 9 places the two departmental orderings side by side over the 166 U.S. departments with at least five publishing faculty.

The two orderings agree far less than the venue sets did. Kendall's τ between them is 0.716, against 0.95 or higher for every pair of venue sets in Table 4; the median department moves 13 positions and one moves 71. Only 6 of the 10 departments in the CSRankings top ten remain in the top ten when measured per head. Figure 2 plots the disagreement directly.

The direction of the change is systematic. CMU leads the CSRankings ordering with 183 publishing faculty and places seventh per head; Stanford places tenth overall and first per head with 61. Georgia Tech is the sharpest case inside the top five: fourth by CSRankings score and thirtieth per head. Northeastern, eleventh overall, is thirty-ninth. Running the other way, Harvard is 39th overall and 9th per head on 38 faculty, and UNC is 43rd overall and 15th per head on 46.

Table 10 lists the departments that gain most from the change of question. They are small and selective, and the volume-weighted ordering buries them.

Neither ordering is more correct. A large department genuinely produces more research, which is what CSRankings sets out to measure, and an applicant choosing by prestige or by breadth of available advisors is right to weigh size. But an applicant asking how productive the people around them will be is asking the per-head question, and the volume-weighted ordering does not answer

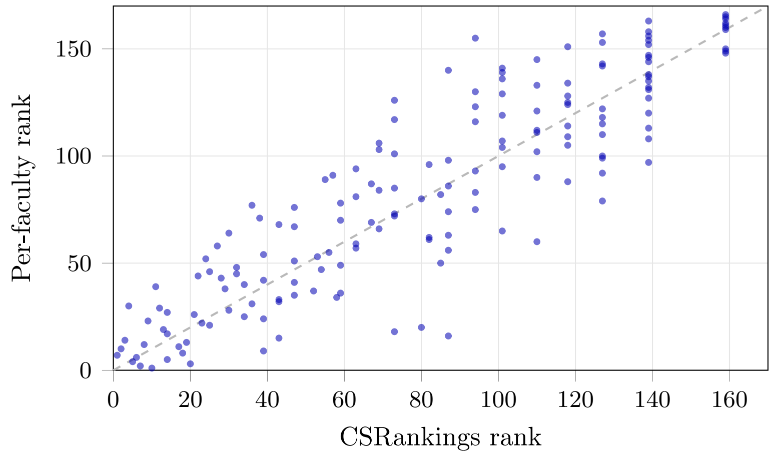


Figure 2: Departmental rank under the CSRankings score against rank by adjusted output per publishing faculty member, for the 166 U.S. departments with at least five publishing faculty. Points below the diagonal do better per head than in aggregate. $\tau = 0.716$.

Table 10: Departments gaining the most positions when ordered per publishing faculty member rather than by CSRankings score.

Department	CSRankings rank	Per-faculty rank	Publishing faculty
TTI Chicago	87	16	15
Caltech	80	20	27
UC Merced	73	18	18
New Hampshire	110	60	8
New Mexico	127	79	8

it. Our point is that the two questions have materially different answers — 13 positions of median disagreement — and that a single ordered list conceals which one it is reporting.

6.5 Many departments rest on very few people

Table 11 reports how many departures would move a department out of its own rank band, for the 51 departments ranked in the top 50 with at least five publishing faculty.

Twelve of the 48 departments leave their band on a *single* departure, and 22 — almost half — on two or fewer. The median is three. Table 12 gives named cases.

Rice, ranked 43rd with 39 publishing faculty, falls out of the top 50 when one person leaves. GMU, 32nd with 54 faculty, needs three. At the other end, CMU and Northeastern never leave their bands within the 15-removal search, and MIT needs five departures to leave the top ten.

Fragility tracks concentration, as it should: departments that leave their band on one departure have a median top-3 share of 22.1%, against 15.4% for those needing four or more. The band medians are not monotonic — the top-50 band has the lowest median at 2 — because a department sitting near the bottom edge of a band is close to leaving it regardless of how its output is distributed.

Read alongside §6.1, this sharpens the picture. A rank is contingent on methodology, moving a median of 17 positions across defensible settings; it is also contingent on a handful of individuals continuing to work where they currently do. For an applicant, a department whose standing rests on one or two people is a genuinely different proposition from one of the same rank whose output is

Table 11: Departures required to leave a rank band, by band. Three departments do not leave their band within the 15-removal search and are excluded from the medians.

Band	Departments	Median departures
Top 10	8	3.5
Top 25	15	5
Top 50	25	2
All resolved	48	3

Table 12: Departures required to leave a rank band, selected departments. ">15" means the department does not leave its band within the search horizon. Top-3 share is the fraction of departmental adjusted output produced by its three most prolific members; the individuals are not identified (§8).

Department	Rank	Band	Publishing faculty	Departures
CMU	1	10	183	>15
MIT	5	10	104	5
Northeastern	11	25	115	>15
GMU	32	50	54	3
Rice	43	50	39	1

spread across dozens, and the rank alone does not distinguish them.

6.6 Individual output is more concentrated than departmental output

Departments are aggregates of people, and the aggregate hides a distribution far more skewed than the departmental one. Table 13 summarizes the 5,987 U.S.-affiliated researchers with any credited output in the 10-year window.

The top decile of researchers produces two fifths of all U.S. output, and the median researcher contributes an adjusted count of 2.19 over ten years — the equivalent of roughly one four-author paper every two years. This is the individual-level counterpart of §6.3, and it is the mechanism behind it: departmental concentration is high because the underlying individual distribution is heavy-tailed, and a department is simply a small sample from it. It also explains why fragility is so often resolved in one or two departures.

Breadth is the other surprise. The median researcher is active in two areas and nearly seven in ten take the majority of their output in a single one, which sits awkwardly beside a departmental score (Eq. 1) that rewards breadth across 27 of them. Departmental breadth is assembled from specialists, not produced by generalists.

Table 14 lists the most prolific individuals.

Two observations. The head of the distribution is entirely AI-adjacent: all fifteen take their primary area in machine learning (seven), computer vision (four), NLP (two), or robotics (two). This is a fact about where publication volume is (§6.7) and not a statement about research importance. And adjusted count is not paper count: Ming-Hsuan Yang's 254 papers exceed those of every researcher ranked above him but one, yet he places eighth on adjusted count, because fractional credit divides a paper among its authors and the two orderings diverge for large-collaboration work. A tool that reported raw paper counts would produce a visibly different list.

Table 13: The distribution of adjusted output and breadth across 5,987 U.S.-affiliated researchers with credited output in a 10-year window. Specialists take at least 60% of their output in one area; generalists are active in at least four.

Quantity	Value
Researchers with credited output	5,987
Median adjusted count	2.19
Median breadth (areas)	2
Share of output from the top 1%	8.6%
Share of output from the top 5%	25.9%
Share of output from the top 10%	40.0%
Active in exactly one area	28.8%
Specialists (≥60% in one area)	69.4%
Generalists (≥4 areas)	29.3%

The most prolific researchers are also, mostly, specialists. The widest researchers are a different set, shown in Table 15.

None of these appear in Table 14, and none is close to the top on volume — twelve areas at an adjusted count of 20.1 is a quarter of the leader's output spread twice as wide. Under Eq. 1 a department values these two profiles very differently: because the geometric mean multiplies $(c + 1)$ over *all* areas, a researcher who supplies the only credit in an otherwise empty area moves the departmental score more than a larger producer in a crowded one. This is the individual-level statement of the same effect that decides the departmental comparison in §6.8.

Table 16 gives the most prolific researcher in each of the 27 subfields, which is the query an applicant with a settled subfield actually wants and which no departmental ordering answers.

The overall-rank column is the point of the table. The leader of a subfield is frequently nowhere near the top of the aggregate researcher ordering: the most prolific U.S. researcher in measurement sits 1376th overall, in operating systems 1191st, in computational biology 837th, and in CS education 808th. Only 6 of the 27 subfield leaders appear in the overall top 100, and the median subfield leader ranks 196th. This is not a statement about those researchers; it is a statement about the aggregate ordering, which is dominated by the volume differences between subfields (§6.7) rather than by standing within one. An applicant who searches a subfield sees the first column; an applicant who reads a global list of researchers sees the last, and the two disagree almost completely. The same caution applies to the ordering in Table 14.

6.7 Subfields differ by two orders of magnitude, and the gap is widening

Table 17 reports every top-level subfield at region-wide scale.

The distribution is extraordinarily skewed (Figure 3). Machine learning alone accounts for 24.0% of all U.S. adjusted output. Together with computer vision, NLP, and AI it accounts for 49.8% — half the measured field. At the other end, computational biology, logic and verification, and operating systems each account for well under 1%. The ratio between the largest and smallest subfield is 51:1.

The growth column shows this is recent and accelerating. Machine learning grew 444% against the preceding decade, NLP 189%, security 164%. Three subfields contracted outright: computational biology by 31%, embedded and real-time systems and economics-and-computation by 27% each. CS education grew 251% from a small base, spreading to 107 departments.

Table 14: The fifteen U.S.-affiliated researchers with the highest adjusted count over a 10-year window. “Share” is the fraction of adjusted output in the researcher’s largest area. All quantities are derived from publication counts CSRankings already publishes; see §8.

#	Researcher	Department	Adj.	Papers	Areas	Primary area	Share
1	Sergey Levine	UC Berkeley	78.9	370	6	Machine Learning	71%
2	Mohit Bansal	UNC	57.6	216	5	NLP	61%
3	Pieter Abbeel	UC Berkeley	53.8	246	8	Machine Learning	66%
4	Heng Huang	UMD	51.3	186	8	Machine Learning	42%
5	Stefano Ermon	Stanford	47.5	201	3	Machine Learning	80%
6	Graham Neubig	CMU	46.5	201	7	NLP	70%
7	David P. Woodruff	CMU	46.0	146	9	Machine Learning	57%
8	Ming-Hsuan Yang	UC Merced	45.5	254	6	Computer Vision	76%
9	Dinesh Manocha	UMD	44.3	210	9	Robotics	47%
10	Daniela Rus	MIT	42.5	201	7	Robotics	74%
11	Quanquan Gu	UCLA	40.3	155	6	Machine Learning	91%
12	Trevor Darrell	UC Berkeley	39.1	210	6	Computer Vision	50%
13	Alan L. Yuille	Johns Hopkins	36.7	193	5	Computer Vision	70%
14	J. Zico Kolter	CMU	35.8	134	4	Machine Learning	87%
15	Kristen Grauman	UT Austin	34.6	107	4	Computer Vision	77%

Table 15: The widest-ranging researchers among the 300 most prolific, by number of top-level areas with credited output, 10-year window.

Researcher	Department	Areas	Adj.	Primary area
Ion Stoica	UC Berkeley	12	20.1	Machine Learning
Yanzhi Wang	Northeastern	12	15.9	Design Automation
Cong Liu	UC Riverside	12	14.1	Embedded & Real-Time
Isil Dillig	UT Austin	12	13.6	Programming Languages
Dawn Song	UC Berkeley	11	25.3	Machine Learning
Xiangyu Zhang	Purdue	10	20.3	Software Engineering
Yiran Chen	Duke	10	18.5	Design Automation
Osbert Bastani	Penn	10	15.3	Machine Learning

This has a direct consequence for how Eq. 1 behaves that is easy to miss. The geometric mean gives every one of the 27 areas equal weight in the exponent, but the areas are not equally sized, so a unit of adjusted output is worth far more to a department’s score in operating systems — where the national total is 147.9 and $\ln(c+1)$ is steeply increasing — than in machine learning, where a department must add enormous volume to move the same factor. A department that hires one systems researcher may gain more score than one that hires three machine-learning researchers. This is arguably the correct incentive for a breadth-rewarding measure, but it is nowhere visible in the output, and it is why per-area credit and not just totals must be shown.

Subfield leadership also turns over. Table 18 lists the subfields whose leading department changed against the preceding period.

CMU now leads 11 of the 27 subfields, having taken NLP, HCI, and algorithms from three different departments while losing databases to MIT. This is the concrete content behind its invariant first place in Table 3: it is not first because it is far ahead anywhere in particular, but because it is at or near the front almost everywhere, which is exactly what Eq. 1 rewards.

Table 16: The most prolific U.S.-affiliated researcher in each top-level subfield, 10-year window, with their adjusted count *within that subfield* and their overall position in the researcher ordering. Subfields are ordered by region-wide output.

Subfield	Researcher	In-area adj.	Department	Overall #
Machine Learning	Sergey Levine	55.8	UC Berkeley	1
Computer Vision	Ming-Hsuan Yang	34.4	UC Merced	8
NLP	Mohit Bansal	34.9	UNC	2
AI	Heng Huang	18.0	UMD	4
Robotics	Daniela Rus	31.6	MIT	10
HCI	Chris Harrison	18.5	CMU	111
Algorithms	Thatchaphol Saranurak	18.3	UMich	117
Security	Zhiqiang Lin	12.6	Ohio State	165
Databases	Immanuel Trummer	15.9	Cornell	160
Architecture	Moinuddin K. Qureshi	11.8	Georgia Tech	236
Info Retrieval	Hamed Zamani	12.2	UMass Amherst	242
Design Automation	David Z. Pan	10.8	UT Austin	300
Software Engineering	Hridesh Rajan	8.2	Tulane	672
Cryptography	Stefano Tessaro	16.0	UW	127
Graphics	Ravi Ramamoorthi	10.3	UCSD	106
Visualization	Kwan-Liu Ma	7.3	UC Davis	580
Programming Languages	Isil Dillig	5.9	UT Austin	259
Mobile Computing	Xinyu Zhang	11.7	UCSD	107
Networks	Mohammad Alizadeh	6.8	MIT	354
HPC	Devesh Tiwari	6.3	Northeastern	318
CS Education	Craig B. Zilles	7.1	UIUC	808
Measurement	Ziv Scully	4.1	Cornell	1376
Embedded & Real-Time	James H. Anderson	9.6	UNC	515
Econ & Computation	S. Matthew Weinberg	7.8	Princeton	196
Operating Systems	Nickolai Zeldovich	4.8	MIT	1191
Logic & Verification	Kuldeep S. Meel	4.9	Georgia Tech	70
Comp. Biology	Carl Kingsford	5.9	CMU	837

Table 17: U.S. output by subfield over a 10-year window, with growth against the equal-length preceding period. “Depts” and “Researchers” count distinct entities with any credit in the subfield; “Top-5” is the share of the subfield’s national output held by its five largest departments.

Subfield	Adjusted	Growth	Depts	Researchers	Per res.	Leader	Top-5
Machine Learning	5382.7	+444%	165	2422	2.22	CMU	24.2%
Computer Vision	2076.2	+83%	156	1185	1.75	CMU	20.4%
NLP	1919.9	+189%	154	1175	1.63	CMU	22.1%
AI	1806.4	+146%	172	1861	0.97	CMU	14.5%
Robotics	1707.6	+52%	144	906	1.89	CMU	24.0%
HCI	1476.3	+146%	146	1202	1.23	CMU	28.5%
Algorithms	1241.7	+20%	91	480	2.59	CMU	30.2%
Security	1137.9	+164%	149	1039	1.10	Georgia Tech	19.3%
Databases	573.2	+49%	125	570	1.01	MIT	20.6%
Architecture	565.2	+82%	101	608	0.93	UIUC	26.7%
Info Retrieval	438.7	+75%	140	729	0.60	UIUC	22.8%
Design Automation	407.5	+10%	118	412	0.99	UCSD	25.4%
Software Engineering	397.0	+60%	118	417	0.95	CMU	25.5%
Cryptography	336.0	+72%	64	160	2.10	UW	32.6%
Graphics	315.6	+10%	73	229	1.38	CMU	36.5%
Visualization	310.4	+76%	110	337	0.92	Utah	23.5%
Programming Languages	309.1	+47%	91	363	0.85	CMU	25.6%
Mobile Computing	301.2	+51%	105	413	0.73	UCSD	22.2%
Networks	290.4	+29%	91	405	0.72	Princeton	32.3%
HPC	282.0	−4%	123	444	0.63	Ohio State	18.6%
CS Education	264.4	+251%	107	336	0.79	UCSD	28.7%
Measurement	220.5	+10%	91	349	0.63	Northeastern	25.5%
Embedded & Real-Time	160.4	−27%	76	164	0.98	WUSTL	38.1%
Econ & Computation	154.4	−27%	55	142	1.09	Cornell	30.6%
Operating Systems	147.9	+85%	65	274	0.54	UC Berkeley	33.5%
Logic & Verification	119.6	+18%	67	159	0.75	Georgia Tech	28.1%
Comp. Biology	105.0	−31%	78	166	0.63	CMU	33.9%

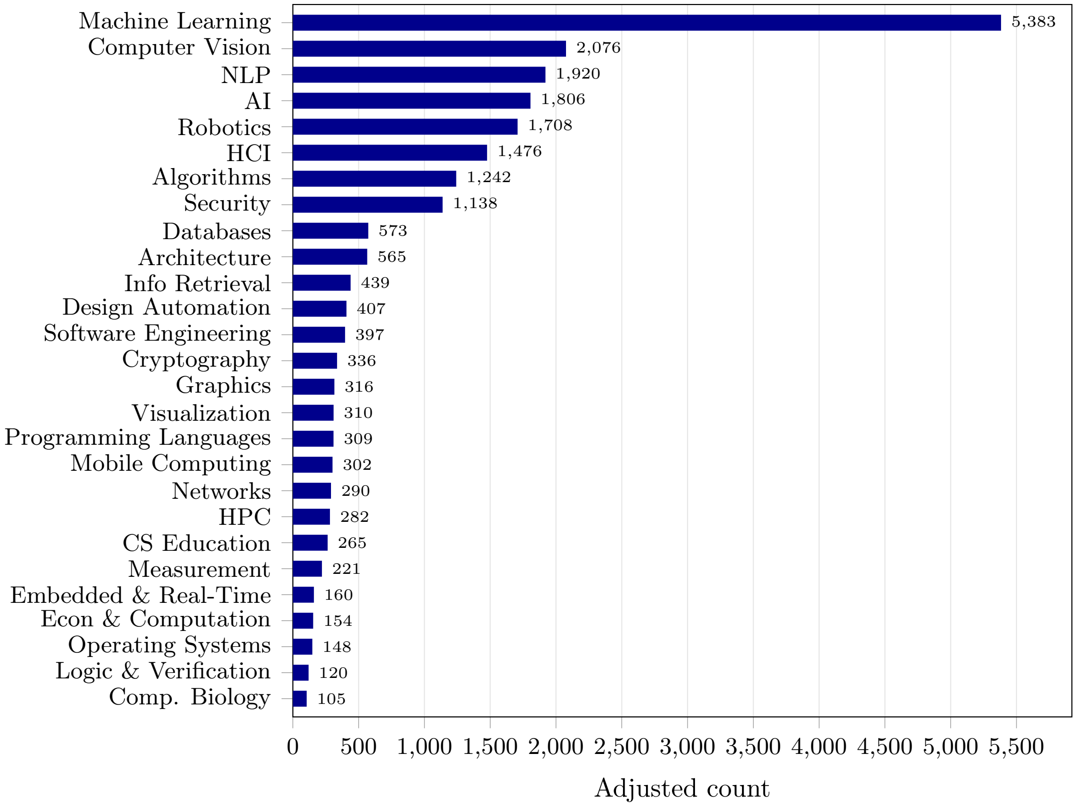


Figure 3: U.S. adjusted publication count by top-level subfield over a 10-year window. Machine learning alone accounts for 24.0% of the total; the four AI-adjacent subfields together account for 49.8%.

Finally, the interface compares two subfields directly. Table 19 gives machine learning against programming languages, chosen as an extreme pair.

The two subfields are separated by a factor of 17 in output and a factor of 6.7 in participation, and 1,857 researchers entered machine learning in the last decade — more than five times the entire current population of programming languages. Yet under Eq. 1 they contribute symmetric factors to a department's score. An applicant told that a department is "ranked 12th" learns nothing about which side of this divide its strength lies on, which is why the per-subfield view (§3.5) rather than the rank is the thing we put in front of them.

6.8 Head-to-head comparison isolates what a rank gap is made of

Table 20 compares UIUC and MIT, three positions apart in the current ranking.

This is the clearest single demonstration of what Eq. 1 actually measures. The two departments' total adjusted output differs by 2.4 — 0.3% — and MIT wins the three largest per-area margins outright, taking machine learning by 71.3, algorithms by 48.5, and robotics by 48.5. Yet UIUC ranks three positions higher.

The decomposition explains it. Every area in the bottom block is one where MIT has almost no credited output: 2.3 in information retrieval, 0.6 in CS education, 0.0 in embedded and real-time systems. Because the score multiplies $(c + 1)$ across all 27 areas, the move from 0.0 to 3.5 contributes more to the product than the move from 195.8 to 267.1 — $\ln(4.5) - \ln(1.0) = 1.50$

Table 18: Subfields whose leading U.S. department changed between the preceding decade and the current one.

Subfield	Former leader	Current leader	Growth
NLP	Stanford	CMU	+189%
HCI	UW	CMU	+146%
Algorithms	MIT	CMU	+20%
Databases	CMU	MIT	+49%
Architecture	UMich	UIUC	+82%
Cryptography	UCLA	UW	+72%

Table 19: Two subfields head to head, 10-year window against the preceding decade. “New researchers” are those with credit in the subfield now and none in the preceding period.

	Machine Learning	Programming Languages
Adjusted count	5382.7	309.1
Preceding decade	989.1	210.0
Growth	+444%	+47%
Departments active	165	91
Researchers active	2422	363
New researchers	1857	206
Leading department	CMU (315.1)	CMU (24.1)
Second	MIT (267.1)	MIT (15.6)
Third	UC Berkeley (264.3)	Penn (14.3)
Researchers active in both	113	

against $\ln(268.1) - \ln(196.8) = 0.31$. The rank gap is made almost entirely of areas where neither department is nationally dominant and one of them is simply absent.

Neither the raw margins nor the rank alone would tell a reader this. A prospective machine learning student reading “UIUC is ranked above MIT” is being told something true about a breadth-weighted aggregate and something false about the thing they care about, where MIT leads by 36%. This is the central case for reporting per-area credit alongside the ordinal.

Table 21 gives a second pair further down the ranking, where the same mechanism operates.

Rice is the stronger machine learning and AI department by a wide margin — 2.6× in machine learning — and ranks eleven positions lower. GMU’s rank advantage is built from HCI, security, visualization, software engineering, and graphics, five areas in which Rice is near zero. Recall from §6.5 that Rice leaves the top 50 on a single departure while GMU needs three: the two comparisons together say that these departments are not close substitutes at any level of detail, despite sitting eleven positions apart in an ordering that reduces both to one number.

The same view works for individuals. Table 22 compares the two most prolific U.S. researchers in the window.

The scoreboard reports a clear leader on volume and a tie on areas led, which is the honest summary: these are researchers in adjacent but different subfields, and the aggregate that puts one above the other is dominated by a single area in which they barely overlap. The interface says so rather than declaring a winner, and this is the reason the comparison view reports areas led

Table 20: UIUC against MIT, 10-year window. "Areas led" counts top-level areas where the department has strictly more adjusted credit.

	UIUC	MIT
Overall rank	2	5
Papers	3488	3294
Adjusted count	782.1	779.7
Publishing faculty	127	104
Areas led	17	10
Largest per-area margins		
Machine Learning	195.8	**267.1**
Algorithms	37.8	**86.3**
Robotics	37.0	**85.5**
NLP	**72.1**	34.3
HCI	**42.6**	22.7
Areas that produced the rank gap (Eq. 5)		
Info Retrieval	**33.1**	2.3
CS Education	**14.5**	0.6
Security	**48.8**	8.8
HPC	**6.4**	0.5
Embedded & Real-Time	**3.5**	0.0

alongside the total instead of collapsing to one verdict.

6.9 Funding orders departments differently from publications

The NSF snapshot provides a second, independently produced signal over the same roster. Table 23 orders U.S. departments by attributed NSF award value over the same 10-year window and reports their publication rank alongside.

The head of the two orderings agrees closely — the same four departments lead both — but agreement decays quickly. Across the 182 matched departments, Kendall's τ between funding rank and publication rank is 0.778, and the median department sits 12 positions apart in the two orderings. That is worse agreement than any pair of venue sets in Table 4 and comparable to the per-faculty disagreement of §6.4. Figure 4 plots it.

The divergent cases are informative in both directions (Table 24). Rice is 43rd on publications and 14th on funding. Texas A&M is 34th on publications and 167th on funding — a 133-position gap, by far the largest in the data, and one worth treating as a question about the matching rather than a finding about the department (§7).

Individual attribution is far more concentrated than departmental attribution, and far more sensitive to the equal-split assumption (Table 25).

The gap between the two money columns is the equal-split assumption made visible. The top investigator's five awards are almost entirely his own on paper ($41.2M attributed against $42.3M of project value), while the second's eight awards carry $109.3M of project value of which the equal split assigns him $21.3M. Neither number is the truth about how much money either person directs, and the interface shows both precisely so that neither can be read as if it were. Note also that this list has almost no overlap with the publication leaders of Table 14: of the twelve, only Yiran Chen

Table 21: GMU against Rice, 10-year window.

	GMU	Rice
Overall rank	32	43
Papers	776	678
Adjusted count	178.7	158.3
Publishing faculty	54	39
Areas led	15	12
Largest per-area margins		
Machine Learning	14.5	**37.8**
AI	9.3	**23.8**
Computer Vision	5.5	**17.3**
HCI	**22.1**	2.1
Security	**22.6**	3.6
NLP	**22.2**	4.4
Areas that produced the rank gap (Eq. 5)		
HCI	**22.1**	2.1
Visualization	**5.3**	0.0
Security	**22.6**	3.6
Software Engineering	**7.7**	0.8
Graphics	**5.5**	0.3

appears anywhere in the breadth or output tables above. Funding and publication volume are close to orthogonal at the individual level.

The award population itself is worth characterizing. Of the 10,226 awards in the window, 81.6% are from the CISE directorate, with education (486), engineering (463), and mathematical and physical sciences (284) making up most of the remainder — so this is not a CISE-only view, and a department strong in interdisciplinary work will show funding that its publication record does not predict. The largest programs by award count are Secure & Trustworthy Cyberspace (929), Software & Hardware Foundations (751), Information Integration & Informatics (676), and Robust Intelligence (588); these are the program names the funding autocomplete offers as its third suggestion group (§5.6).

Finally, collaborative awards are the reason the interface reports project value separately at all. The largest matched collaboration in the window is a $12.0M National AI Research Resource award split across 7 institutional portions, followed by four $10.0M Expeditions-in-Computing collaborations spanning 4 to 10 portions each. Attributed per investigator, these become unremarkable figures; read as projects, they are among the largest things in the data.

7 Limitations

The proxy is upstream of everything. CS Picks inherits CSRankings's premise that fractional publication credit at selective conferences is a useful proxy for research activity. Work published in journals, at workshops, or at venues outside the curated lists is invisible, as is teaching, mentorship, software, and service. Our stability analysis quantifies sensitivity *within* this framework; it says nothing about the framework's validity.

Table 22: Two researchers head to head, 10-year window.

	Sergey Levine	Mohit Bansal
Department	UC Berkeley	UNC
Papers	370	216
Adjusted count	78.9	57.6
Active areas	6	5
Areas led	3	3
Machine Learning	**55.8**	11.2
NLP	0.8	**34.9**
Robotics	**19.8**	0.5
Computer Vision	1.5	**5.8**
AI	0.3	**5.2**

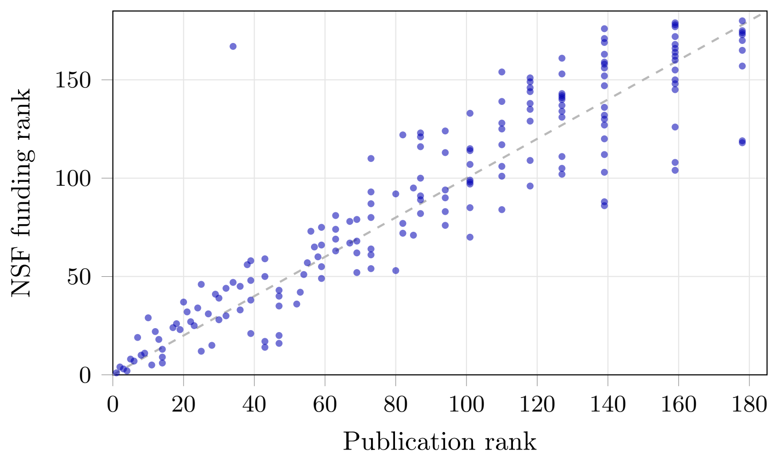


Figure 4: Publication rank against attributed-NSF-funding rank for the 182 matched U.S. departments, 10-year window. $\tau = 0.778$, median displacement 12 positions.

Aggregated inputs. Because the publication table carries no author order or paper identity, alternative credit models — first-author, last-author-weighted, harmonic — cannot be evaluated at corpus scale without per-author DBLP retrieval. We regard this as the most interesting blocked direction.

Historical attribution quality. OpenAlex affiliation histories contain errors, which is why a community-corrections file is merged over them. Results computed under historical attribution should be treated as provisional.

Fragility is a counterfactual. The analysis assumes a department's remaining output is unchanged when someone leaves, which no real department experiences: positions are refilled, collaborations are redistributed, and students continue. It measures how concentrated current output is, expressed in units of departures, not a forecast.

Name identity. All joins are by name. CSRankings disambiguates with numeric suffixes, which we preserve, but coauthor matching against DBLP remains exact-match by construction: we prefer

Table 23: The fifteen U.S. departments with the largest attributed NSF award value, awards dated 2016–2026. "Attributed" splits each award equally among its listed investigators (§3.10). "Pub. rank" is the CSRankings rank over the same window.

#	Department	Attributed	Awards	Funded faculty	Pub. rank	Δ
1	CMU	\$142.7M	293	124	1	0
2	Georgia Tech	\$119.9M	278	103	4	+2
3	UCSD	\$119.1M	235	97	3	0
4	UIUC	\$108.3M	236	96	2	−2
5	Northeastern	\$107.0M	216	75	11	+6
6	UT Austin	\$91.4M	109	40	14	+8
7	UW	\$82.3M	171	58	6	−1
8	MIT	\$78.5M	173	69	5	−3
9	UW–Madison	\$75.8M	157	58	14	+5
10	UMD	\$75.5M	199	78	8	−2
11	UMich	\$73.2M	179	69	9	−2
12	Northwestern	\$72.5M	153	54	25	+13
13	NYU	\$70.8M	166	55	14	+1
14	Rice	\$63.0M	83	28	43	+29
15	Virginia	\$62.3M	151	47	28	+13

to miss a match than to attribute one researcher's collaborators to another.

Funding attribution is a lower bound, and an uneven one. Three distinct effects bound the funding view. The equal split among listed investigators is an assumption NSF's data cannot confirm or refute. Awards are retrieved per current roster member and matched against their *current* affiliation, so a grant held at a previous institution, or by a departed or retired investigator, is dropped rather than reassigned — which makes every departmental total a lower bound whose tightness varies with faculty turnover. And a PI whose name resolution fails is silently absent. The Texas A&M case in Table 24 — 34th on publications, 167th on funding — is large enough that we would treat it as a matching artifact to be investigated rather than a substantive finding, and we report it as such rather than omitting it.

Subfield boundaries are inherited, not derived. The 27 top-level areas and the venue-to-area assignment come from CSRankings. Results in §6.7 about subfield size and growth are therefore statements about that taxonomy: a venue reassigned upstream moves output between subfields, and work at a venue assigned to no area is counted nowhere. The striking machine-learning growth figure in particular reflects both genuine growth and the expansion of what the taxonomy counts as machine learning.

Point-in-time data. All results are computed against upstream data as fetched, and CSRankings updates continuously. Every number here is regenerated by a single script, so figures can be refreshed, but any specific value in this paper should be read as of its generation date rather than as a stable property.

Table 24: Departments where the funding and publication orderings disagree most, in each direction. Gap is publication rank minus funding rank.

Department	Funding rank	Pub. rank	Gap
Funded well above publication standing			
Rhode Island	104	159	+55
New Mexico State	86	139	+53
Missouri	88	139	+51
UNLV	108	159	+51
Hawaii at Manoa	103	139	+36
Publishing well above funding standing			
Texas A&M	167	34	−133
Brigham Young	154	110	−44
Arizona	122	82	−40
UC Merced	110	73	−37
Old Dominion	176	139	−37

8 Ethical considerations

Rankings applied to individuals invite misuse in ways that rankings applied to institutions do not. We adopt several constraints. CS Picks presents no individual-level metric as an evaluation, and reports faculty output only as the publication counts its sources already make public. Every derived quantity is labeled with what it measures and what it cannot, inline rather than in documentation. The system infers no protected characteristics. Because it is entirely client-side, it collects no record of queries.

Why some individuals are named and others are not. The researcher tables of §6.6 and the funding table of §6.9 name people; the fragility analysis of §3.7 does not, and the line between them is deliberate. Tables 14–16 report publication counts that CSRankings and DBLP already publish per person, in the same units, and the tool displays them because an applicant looking for an advisor needs exactly this. The fragility procedure produces something different in kind: a ranked list of the individuals whose departure would most damage their employer, which exists nowhere in the public record and which invites personnel conclusions this work should not support. Its analytic content — how concentrated a department's output is — survives fully in the aggregate, so the names are withheld with nothing lost. The concentration results of §6.3 are reported distributionally for the same reason.

Ordering people is more hazardous than ordering departments. Table 16 is included partly as its own counterargument: the median subfield leader ranks 196th on the aggregate ordering and the leader in measurement ranks 1376th, which demonstrates that a global ranking of researchers is mostly a ranking of subfield publication volume. We report the aggregate ordering only alongside the per-subfield one, and the interface surfaces researchers through search and subfield context rather than as a leaderboard.

Table 25: The twelve investigators with the largest attributed NSF share, 2016–2026. "Project value" is the full value of the awards they participate in, which for large collaborative awards is many times the attributed share.

#	Investigator	Department	Attributed	Project value	Awards
1	Adam R. Klivans	UT Austin	$41.2M	$42.3M	5
2	Richard G. Baraniuk	Rice	$21.3M	$109.3M	8
3	Sidney K. D'Mello	Colorado Boulder	$11.0M	$47.1M	10
4	Ashok K. Goel	Georgia Tech	$10.8M	$20.8M	4
5	John Preskill	Caltech	$10.7M	$25.3M	2
6	James C. Lester	NC State	$10.4M	$38.1M	11
7	Ellie Pavlick	Brown	$10.3M	$21.0M	2
8	Suresh Venkatasubramanian	Brown	$10.0M	$20.0M	1
9	Tamara Sumner	Colorado Boulder	$9.4M	$47.9M	7
10	David R. Choffnes	Northeastern	$8.9M	$22.2M	12
11	Tiffany Barnes	NC State	$7.4M	$23.3M	18
12	Yiran Chen	Duke	$7.2M	$30.2M	20

9 Conclusion

CS Picks began as a companion to a guide for prospective doctoral students [7] and became an argument about presentation. The computation underlying CSRankings is transparent and reproducible; what a single ordered list cannot convey is how contingent its output is on choices that were made silently. Once the ranking pipeline is fast enough to run many times — roughly 100 ms in a browser, in our implementation — that contingency becomes directly measurable and directly displayable.

The measurement yields a finding we did not anticipate: rank stability is sharply stratified. The top ten departments are nearly invariant while the median department moves 17 positions across defensible settings, which means the ranking is most reliable in the region where readers need it least.

Reporting at three levels sharpens rather than softens that conclusion. The per-faculty ordering disagrees with the CSRankings ordering by a median of 13 positions and an NSF-funding ordering by 12, so the departmental number is one of several defensible aggregates rather than the aggregate. Half of all measured output now sits in four AI-adjacent subfields, so a departmental total is dominated by a composition question the ordinal never states. And the top decile of individuals produces 40% of all output, so a department is a small sample from a heavy-tailed distribution — which is why a single departure resolves a quarter of the fragility cases we examined.

We believe the appropriate response is not to propose a better ranking, but to report the envelope alongside the number, and to supply the more local information — which subfield, which faculty, which funding — that the decision actually turns on.

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